Superpixel-Based Brownian Descriptor for Hyperspectral Image Classification

Shuzhen Zhang, *Member, IEEE*, Ting Lu, *Member, IEEE*, Shutao Li, *Fellow, IEEE*, and Wei Fu, *Member, IEEE*

***Abstract*— Exploring effective spectral–spatial feature extraction methods is one of the most focused problems in current hyperspectral image (HSI) classification research. However, complex spectral–spatial structure characteristics in HSIs, e.g., shape-variable spatial structure and nonlinear spectral structure, are difficult to be effectively extracted and jointly represented. To overcome this issue, a novel superpixel-based Brownian descriptor (SBD) method for HSI classification is proposed in this article. In specific, superpixel segmentation is first used to extract shape-adaptive spatial structure information from dimension-reduced HSI, leading to generate nonoverlapping homogeneous 3-D image blocks. Then, similar pixels within the 3-D image block are jointly represented by a new local spectral–spatial feature based on the Brownian descriptor (BD). This is the first time that the BD is introduced to measure both linear and nonlinear correlations among different spectral bands in HSI. On one hand, the integration of superpixel and BD helps to provide much richer and more valuable information for better discrimination between different categories. On the other hand, the SBD can effectively represent the internal structure characters within each 3-D image block of different spatial shapes by a symmetric positive definite matrix of a united form. Finally, considering that the SBD lies on the Riemannian manifold space, a log-Euclidean kernel sparse representation (LKSR) classifier is introduced to obtain the classification results. Experimental results on three widely used real hyperspectral datasets indicate the performance superiority of the proposed SBD method over several state-of-the-art techniques.**

***Index Terms*— Feature extraction, hyperspectral image (HSI) classification, log-Euclidean kernel sparse representation (LKSR), superpixel-based Brownian descriptor (SBD).**

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Shuzhen Zhang is with the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Hunan University, Changsha 410082, China, and also with the College of Information Science and Engineering, Jishou University, Jishou 416000, China (e-mail: shuzhen_zhang@hnu.edu.cn).

Ting Lu and Shutao Li are with the College of Electrical and Information Engineering and the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Hunan University, Changsha 410082, China (e-mail: tingluhnu@gmail.com; shutao_li@hnu.edu.cn).

Wei Fu is with the College of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China (e-mail: fuweihunandaxue@gmail.com).

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I. Introduction

HYPERSPECTRAL image (HSI) can provide rich spectral–spatial information, which has been increasingly applied in different remote sensing fields, such as environmental management, precision agriculture, and military reconnaissance [1]–[3]. HSI classification is an important task in the remote sensing application area, which assigns each pixel with a label to indicate its belonging category. However, both the Hughes phenomenon [4] caused by the high-dimensional characteristic of HSI and the complex spectral–spatial structure characteristics make accurate HSI classification a tough task [5]–[9]. Exploring feature extraction methods is an effective way to improve classification accuracies, which has been raised as a hot research topic.

In the last decades, various feature extraction methods have been proposed for HSI classification. Traditionally, feature extraction methods were mainly developed in the spectral domain, such as the principal component analysis (PCA) [10], the independent component analysis [11], the maximum noise fraction (MNF) [12], and the linear discriminant analysis [13]. However, these extracted spectral-based features are susceptible to uncertain factors, such as noise and light. Considering that the spatial distribution of ground materials is usually locally continuous, various spectral–spatial feature extraction techniques have been introduced to the HSI. For example, in [14] and [15], the Gabor filter with multiple orientations and frequencies is used to exploit the texture information of HSI. In [15], as a commonly used feature descriptor, the local binary pattern (LBP) filter is applied to enhance the HSI classification performance. In [16], the extended morphological profiles' filter, which is constructed by sequentially performing opening and closing operations on the image, is utilized to extract the structure information for the remote sensing image. In [17], the spatial information of the HSI is exploited by the extended morphological attribute profile (EMAP) filter that can characterize the spatial structure of objects according to different types of attributes. In [14] and [18], an edge-preserving filter, which can smooth out the image textures without blurring edges, is utilized to combine the spatial and spectral information of HSI. However, these feature extraction methods attempt to excavate and represent spatial information within the window-based local region of fixed shape or fixed scale. To excavate more discriminative features for HSI classification from adaptive-shape homogeneous regions, the superpixel segmentation technique has been widely researched [19]–[22].

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Generally, each superpixel is an adaptive segmentation region that consists of a series of adjacent pixels according to the spatial structure. In the superpixel-based HSI classification methods, pixels in a superpixel are generally assumed to be the same class and can be simultaneously dealt with. However, for the irregularity of the spatial shape, most of the proposed feature extraction methods are not suitable for the superpixel. Hence, it is necessary to explore new feature extraction methods, which can provide the discriminative information based on the irregular-shape superpixel for the HSI classification.

Recently, the correlation feature has drawn much attention in various computer vision tasks [23]–[26], which has also been applied in processing HSI classification tasks [27]–[31]. In [27] and [28], correlation coefficients among training and test samples are calculated for HSI classification. In [29]–[31], considering that different ground materials have different correlations among spectral bands, the local covariance matrix is adopted to exploit the spectral correlation information for HSI classification. The local covariance matrix representation (LCMR) can capture second-order statistics of the local spatial–spectral signal and is independent of the size of the local region. However, the classical correlation feature only measures linear dependence among spectral bands, which cannot characterize the nonlinear correlation actually existed among spectral bands in HSI. Székely and Rizzo [32] introduced the Brownian covariance, which provides a new method to measure dependence and test the joint independence of random vectors in arbitrary dimensions. In contrast to the classical covariance, the Brownian covariance can not only measure the degree of the linear relationship between two real-valued random variables but also capture nonlinear and nonmonotone dependence between the two real-valued variables.

In this article, a novel superpixel-based Brownian descriptor (SBD) method is proposed for HSI classification. In the proposed SBD method, the shape-adaptive spatial information and rich spectral correlation information of different ground objectives are captured to construct the discriminative spectral–spatial feature for HSI classification. Specifically, on a dimension-reduced HSI, the superpixel segmentation technique is first used to generate a homogeneous 3-D image block. Then, the BD is utilized on each image block to extract both linear and nonlinear spectral correlation information. Finally, considering that the obtained SBD is a symmetric positive definite (SPD) matrix and lies on a Riemannian manifold space, the SBD is input a log-Euclidean kernel sparse representation (LKSR) classifier [23] to yield the classification result. In the proposed method, the BD is the first time utilized to characterize the complex interspectral correlation information in the HSI. Moreover, since the BD is shape-independent, it can be employed on the superpixel-based 3-D image block with different sizes. In addition, the integration of BD and superpixel can more reasonably utilize spatial information to extract more discriminative spatial–spectral features for HSI classification.

The remainder of this article is structured as follows. Section II briefly overviews the related work including Brownian covariance and entropy rate superpixel (ERS) segmentation method [33]. Section III introduces the proposed SBD method for HSI classification. In Section IV, experimental results and analysis are reported. Conclusions and future research directions are given in Section V.

II. Related Works

A. Brownian Covariance

The Brownian motion is a stochastic process for describing the random movement of suspended particles impinged on by molecules. The Brownian covariance is a mathematical statistic to reflect the correlation between random particles of the process. Székely and Rizzo [32] pointed that the Brownian covariance $\mathcal{W}$ coincides with the distance covariance $\mathcal{V}$. That is, for arbitrary $\mathbf{x} \in \mathbb{R}^k$ and $\mathbf{y} \in \mathbb{R}^f$ with finite second moments,

$$\mathcal{W}(\mathbf{x}, \mathbf{y}) = \mathcal{V}(\mathbf{x}, \mathbf{y}). \tag{1}$$

Compared to the classical covariance, the Brownian covariance can be computed between $\mathbf{x}$ and $\mathbf{y}$ in arbitrary dimensions. Furthermore, it also can engrave the nonlinear correlation between variables, which is a natural extension of classical covariance. Moreover, an important property of the Brownian covariance is that $\mathcal{W}(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x}$ and $\mathbf{y}$ are independent.

To calculate the Brownian covariance, the sample distance covariance $\mathcal{V}_L$ will be used. Let $(\boldsymbol{X}, \boldsymbol{Y}) = \{(\mathbf{x}_l, \mathbf{y}_l) : l = 1, 2, \ldots, L\}$ be an observed random sample from the joint distribution of random vectors $\mathbf{x} \in \mathbb{R}^k$ and $\mathbf{y} \in \mathbb{R}^f$. If the mathematical expectations are $E|\mathbf{x}|_k < \infty$ and $E|\mathbf{y}|_f < \infty$, then almost surely

$$\lim_{L \to \infty} \mathcal{V}_L(\boldsymbol{X}, \boldsymbol{Y}) = \mathcal{V}(\mathbf{x}, \mathbf{y}). \tag{2}$$

Therefore, from formulas (1) and (2), the Brownian covariance can be obtained by calculating the sample distance covariance. The specific calculation process will be described in Section III-B.

B. Entropy Rate Superpixel Segmentation

Superpixel segmentation is used to adaptively segment a whole image into multiple nonoverlapping regions, where one region corresponds to one superpixel. Recently, several popular superpixel segmentation methods have been developed [22], [33]. Among them, the graph-based ERS segmentation method has been demonstrated to be very effective. Specifically, an image is first mapped to a graph $G_I = (G_v, G_e)$, in which G_v is the vertex set that represents the pixels of the image and G_e is the edge set that denotes the pairwise similarities between adjacent pixels. In ERS, a subset of edges $G_s \subseteq G_e$ is generated by solving a graph optimizing problem as follows:

$$\max_{G_s} \left\{ R_g(G_s) + \alpha \Gamma_g(G_s) \right\}, \quad \text{s.t. } G_s \subseteq G_e \tag{3}$$

where $R_g(G_s)$ is an entropy rate term of the random walk on graph G_I to obtain the compact and homogeneous superpixels, $\Gamma_g(G_s)$ is a balancing term to enable superpixels with similar sizes, and $\alpha \geq 0$ is the weight of the balancing term. In [33], a greedy method is adopted to effectively solve this problem.

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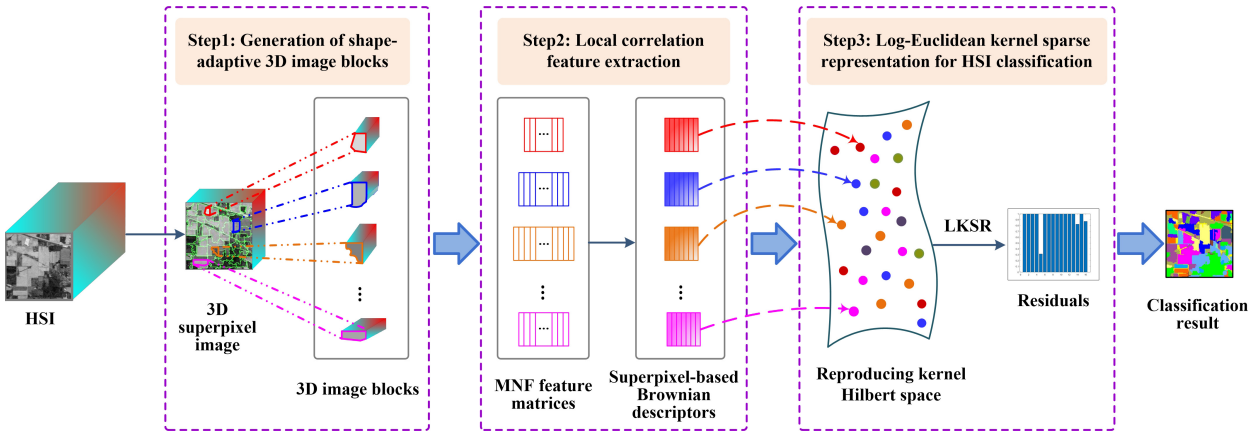


Fig. 1. Schematic of the proposed SBD method for HSI classification. The colored dotted lines from the second step to the third step indicate the kernel mappings that map the SBDs into an RKHS.

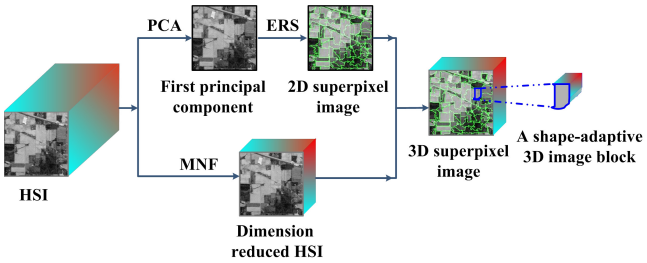


Fig. 2. Illustration of generating a shape-adaptive 3-D image block in HSI.

III. Proposed Method

Considering the complex spectral–spatial characteristics in HSI, i.e., variable spatial structure and nonlinear spectral correlation, a novel SBD is proposed to extract and represent discriminative spectral–spatial information. The proposed SBD method for HSI classification mainly consists of three parts: 1) generation of shape-adaptive 3-D image blocks; 2) local correlation feature extraction for the block; and 3) LKSR classifier with the local correlation feature for HSI classification. The schematic of the proposed method is demonstrated in Fig. 1, and a detailed description of the method is given in the following.

A. Generation of Shape-Adaptive 3-D Image Blocks

In HSI, neighboring pixels always present high spatial-correlation and similar spectral characteristics; they should belong to the same object category. Therefore, it is reasonable to group neighboring pixels to extract local spatial information. To achieve this goal, shape-adaptive homogeneous 3-D image blocks based on superpixel segmentation are first generated here. Instead of the window-based region of fixed shape or fixed size, superpixel segmentation can help to extract richer spatial information for better discrimination performance. The illustration of generating shape-adaptive a 3-D image block is shown in Fig. 2.

As shown in Fig. 2, a dimension reduced HSI is divided into multiples shape-adaptive 3-D blocks based on the superpixel segmentation technique. Specifically, the PCA method is first used on the original HSI to obtain the first principal component, which contains the major information of the HSI. Then, a 2-D superpixel image for the first principal component is gained by adopting ERS segmentation method. At the same time, the MNF method, which aims to find a projection matrix to maximize the signal-to-noise ratio of transformed data, is applied on the original HSI to discard noise and produce a dimension reduced HSI [12]. After that, the 2-D superpixel image combines the dimension reduced HSI to generate a 3-D image marked by superpixels. In the 3-D image, pixels in one superpixel are constructed as a shape-adaptive 3-D image block. In this proposed method, for a training sample, the 3-D image block containing the training sample is the corresponding training block, while any 3-D image block that is removed from all training samples is a test block.

To obtain the homogeneous block, the determination of the superpixel number before the superpixel segmentation is very important. Here, an estimation method of the superpixel number N_{supp} is given. Generally, the more complicated the spatial construction of an image is, the larger number of superpixels should be set. Meanwhile, the structural complexity of an image can be reflected by the edge of the image. Therefore, for the HSI, the Canny filter [34] is first applied on the first principal component of the HSI to obtain the edge image. The number of the nonzero pixels in the edge image, which reflects the spatial complexity of the HSI, is counted and noted as N_{edge}. Then, the number of superpixels N_{supp} is defined as

$$N_{\text{supp}} = \eta \times N_{\text{edge}} \tag{4}$$

where η is the empirical coefficient.

In addition, since high redundancy exists in each block, T_s most similar pixels within a block will be selected to represent the block, which still can provide sufficient spectral information for effective classification. Here, the cosine distance C_d, which is usually used to measure the spectral similarity of two spectral pixels, is adopted [35]. For the training block, C_d is used to calculate the similarity between the training pixel and neighborhood pixels in the training block. For the test block, C_d is utilized to compute the similarity between the mean of all spectral pixels within the block and each spectral pixel

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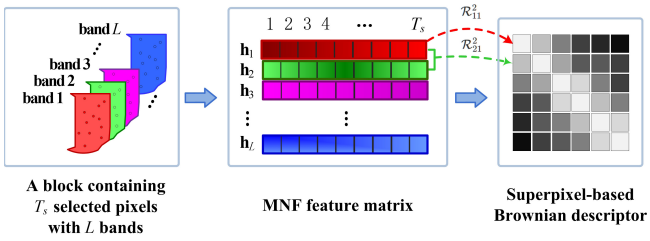


Fig. 3. Illustration of the generation of an SBD.

in the block. According to C_d, T_s most similar pixels in one block are picked out to represent the block.

B. Local Correlation Feature Extraction

In addition to the spatial-correlation information, spectral-correlation information also plays an important role in distinguishing different categories of land covers. Due to the effect of hyperspatial imaging mechanism and external factors, different bands have both linear correlation and nonlinear correlation. However, most existing correlation-based HSI classification methods only consider linear spectral correlation, which cannot well characterize the complex spectral structures characteristics. To solve this problem, the BD is introduced in HSI classification for the first time. The efficient calculation of BD is carried out in accordance with [36], which can fully excavate dependencies among features.

Fig. 3 illustrates the generation of an SBD, which is obtained by calculating of the BD of a shape-adaptive 3-D image block. Specifically, let L denote the band number of the dimension reduced HSI, i.e., the number of MNF components. Let $\boldsymbol{H} = \{\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_L\}$ indicate the L bands of T_s selected pixels within a block. Therefore, each band $\mathbf{h}_l$ $(l = 1, 2, \ldots, L)$ can be treated as a point in a T_s-dimensional space. Formulas for computing the sample distance covariance are employed to calculate the SBD [32], [36]. First, the Euclidean distance between any two bands of the T_s pixels is represented as

$$a_{\ln} = \left|\mathbf{h}_l - \mathbf{h}_n\right|, \quad l, n = 1, 2, \ldots, L. \tag{5}$$

Then, define

$$A_{\ln} = a_{\ln} - \overline{a}_{l\cdot} - \overline{a}_{\cdot n} + \overline{a}_{\cdot\cdot}, \quad l, n = 1, 2, \ldots, L \tag{6}$$

where

$$\overline{a}_{l\cdot} = \frac{1}{L}\sum_{n=1}^{L} a_{\ln}, \quad \overline{a}_{\cdot n} = \frac{1}{L}\sum_{l=1}^{L} a_{\ln}, \quad \overline{a}_{\cdot\cdot} = \frac{1}{L^2}\sum_{l=1}^{L}\sum_{n=1}^{L} a_{\ln}. \tag{7}$$

After that, the nonnegative sample distance covariance $\mathcal{V}_L$ is gained as

$$\mathcal{V}_L^2(\boldsymbol{H}) = \mathcal{V}_L^2(\boldsymbol{H}, \boldsymbol{H}) = \frac{1}{L^2}\sum_{l=1}^{L}\sum_{n=1}^{L} A_{\ln}^2. \tag{8}$$

According to Bak *et al.* [36], the element in the lth row and the nth column of the SBD matrix is defined as

$$\mathcal{R}_{\ln}^2 = \frac{A_{\ln}^2}{\mathcal{V}_L^2(\boldsymbol{H})}. \tag{9}$$

Similar to the classical covariance matrix (CCM), the SBD is also an SPD matrix. The difference between CCM and SBD is that the CCM estimates correlation based on the mean of each band, while the SBD is based on distances between any two bands. After this step, the richer and more discriminative spectral–spatial information within each shape-adaptive 3-D image block can be effectively extracted and jointly represented by a novel feature descriptor. On one hand, effective extraction of spectral–spatial structure information by the introduced SBD can help to more accurately discriminate different classes. On the other hand, regardless of variable spatial shape and the different pixel number of each 3-D block, the feature descriptor can be mathematically represented by an SPD of united form.

C. Log-Euclidean Kernel Sparse Representation for HSI Classification

After the step of local correlation feature extraction, extracted spectral–spatial feature is represented by an SPD matrix, which lies on a Riemannian manifold space, instead of the Euclidean space. Thus, the conventional mathematical model on the Euclidean space cannot be directly applied to the SPD matrix. To address this problem, a kernel sparse representation classification (KSRC) based on a log-Euclidean framework [23], i.e., LKSR classifier, is applied.

In the log-Euclidean framework, a family of kernel functions has been developed, which satisfies Mercer's condition under broad conditions and can effectively depict the geodesic distance between SPD matrices. Among these kernel functions, the log-Euclidean Gaussian kernel is introduced in the LKSR classifier. The log-Euclidean Gaussian kernel is defined as

$$k_e(\mathbf{Z}_1, \mathbf{Z}_2) = \exp\left\{-\gamma \parallel \texttt{logm}(\mathbf{Z}_1) - \texttt{logm}(\mathbf{Z}_2) \parallel_F^2\right\} \tag{10}$$

where $\mathbf{Z}_1$ and $\mathbf{Z}_2$ are two samples in the Riemannian manifold. $\gamma > 0$ is a kernel parameter. $\|\cdot\|_F$ denotes the Frobenius norm, and $\texttt{logm}(\cdot)$ denotes the matrix logarithm operator [37]. For the LKSR method, the kernel sparse coding is conducted in a reproducing kernel Hilbert space (RKHS). A detailed description of the LKSR method is given in the following.

Let $\mathbf{D}_w^m$ $(m = 1, 2, \ldots, M)$ represent SBDs of training samples, where M is the number of the training samples. Let $\mathbf{Z}_w$ be the SBD of a test sample. Similar to the KSRC [38], the LKSR model with SBD on RKHS is formulated as

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\phi(\mathbf{Z}_w) - \phi(\mathfrak{D})\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1 \tag{11}$$

where $\mathfrak{D} = \{\mathbf{D}_w^m\}_{m=1}^M$ is a dictionary on a Riemannian manifold consisting of M atoms $\mathbf{D}_w^m$, ϕ is a function that maps the SPD matrices into an RKHS, $\hat{\boldsymbol{\beta}}$ is the estimated vector of sparse coefficients with respect to the mapping dictionary $\phi(\mathfrak{D})$, and $\lambda > 0$ is a scalar regularization parameter. The SR in (11) can be expanded as

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \; k_e(\mathbf{Z}_w, \mathbf{Z}_w) - 2\boldsymbol{\beta}^T\mathbf{k}_e(\mathbf{Z}_w, \mathfrak{D}) + \boldsymbol{\beta}^T\mathbf{K}_e(\mathfrak{D}, \mathfrak{D})\boldsymbol{\beta} + \lambda\|\boldsymbol{\beta}\|_1 \tag{12}$$

where $\mathbf{k}_e(\mathbf{Z}_w, \mathfrak{D})$ is an $M \times 1$ relation coefficient and $\mathbf{K}_e(\mathfrak{D}, \mathfrak{D})$ is a Gram matrix with size of $M \times M$. A kernel sparse coding

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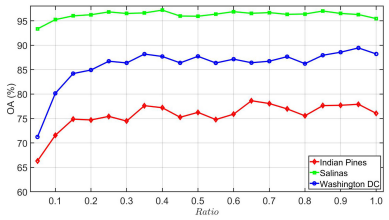


Fig. 4. Sensitivity analysis of parameters Ratio in the proposed SBD method for the three datasets.

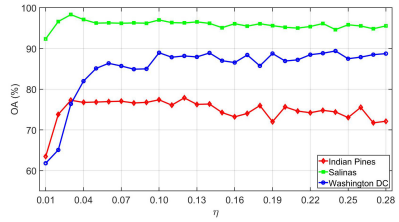


Fig. 5. Effect of the coefficient η to compute the number of superpixels on the proposed SBD method for the three datasets.

method introduced in [39] is adopted to solve the optimization problem in (12).

After obtaining the kernel sparse coefficient $\hat{\boldsymbol{\beta}}$, the class label of the test block $\mathbf{Z}_w$ is determined by the minimum reconstruction residual in the RKHS, which is equivalently expressed as

$$\arg\min_{c=1,2,\ldots,N_C} \delta_c^T(\hat{\boldsymbol{\beta}})\mathbf{K}_e(\mathfrak{D},\mathfrak{D})\delta_c(\hat{\boldsymbol{\beta}}) - 2\delta_c^T(\hat{\boldsymbol{\beta}})\mathbf{k}_e(\mathbf{Z}_w,\mathfrak{D}) \quad (13)$$

where $\delta_c(\cdot)$ is the indicator operation that chooses coefficients related with the c-class and sets the rest coefficients to zero [38]. N_C denotes the number of classes.

IV. Experimental Results and Discussion

In this section, three widely used hyperspectral datasets are used in the experiments, which contains different types of scenes and various land covers. To demonstrate the effectiveness and superiority of the proposed SBD method,[1] both classical and state-of-the-art HSI classification methods are compared, including the support vector machine (SVM) [5], multiple kernel learning for probabilistic outputs of the kernel collaborative representation classification (MPKCRC) [40], EMAP [17], LBP [15], multiscale superpixel-based sparse representation (MSSR) [19], superpixelwise PCA (SuperPCA) [20], correlation coefficient with joint sparse representation (CCJSR) [27], and LCMR [29]. In order to objectively compare the classification performance of different methods, the overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) are adopted as quality indexes. In the following, both visual classification maps and objective metrics comparison results are given.

A. Datasets

In our experiments, three hyperspectral datasets, i.e., the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) Indian Pines image,[2] the Hyperspectral Digital Image Collection Experiment (HYDICE) Washington DC image,[3] and the AVIRIS Salinas image,[2] are used.

1) Indian Pines Image: The Indian Pines image was obtained by the AVIRIS sensor over the agricultural Indian Pines site in Northwestern Indiana. The raw image provides 220 spectral channels in the spectrum from 0.4 to 2.5 μm, where 20 water absorption bands are discarded. The spatial size of this image is 145 $\times$ 145 with a spatial resolution of 20 m per pixel. There are 10 249 labeled pixels with 16 classes in the image. More detailed information on the number of training and test samples is summarized in Table I.

2) Washington DC Image: The Washington DC image was recorded by the HYDICE sensor over the Washington DC Mall. The image consists of 280 $\times$ 307 pixels and 210 spectral bands. The spectral coverage ranges from 0.4 to 2.5 μm, and the spatial resolution of the image is 3 m per pixel. There are six different land-cover classes. The number of training and test samples is listed in Table I.

3) Salinas Image: The Salinas image was captured by the AVIRIS sensor over the area of Salinas Valley, California, USA. The image is of size 512 $\times$ 217 $\times$ 224, where 20 water absorption spectral bands are removed. The geometric resolution of this image is 3.7 m. There are 16 classes, and the detailed number of training and test samples is shown in Table I.

B. Parameter Analysis

For the proposed SBD method, the parameters are set as shown in Table II. Among these parameters, the ratio of the MNF component number to the original spectral band number (i.e., Ratio) is adopted to calculate the band number of dimension reduced HSI. The parameters listed in Table II will be further analyzed as follows.

First, the effect of the band number of dimension reduced HSI, i.e., Ratio, is discussed. Here, the parameter Ratio ranges from 0.05 to 1.0 with a step of 0.05; meanwhile, the other parameters are set as values listed in Table II. As can be seen from Fig. 4, with the increase in Ratio, the OA values on the three datasets increase at first and then present a higher and more stable trend. Since the too small value of Ratio means that fewer bands have remained in the dimension reduced image, that is, more spectral information is discarded, as a result, the discrimination between different land covers will be descended, leading to a lower OA. For the three datasets of different scenes, the best OA values are obtained with various parameter value of Ratio, i.e., Ratio = 0.65 for the Indian Pines dataset, Ratio = 0.95 for the Washington DC data, and Ratio = 0.4 for the Salinas dataset. Considering the overall classification performance and practicality, Ratio = 0.65 is selected for all the remaining experiments.

[1] https://github.com/ShuzhenZhang2019/SBDFE

[2] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

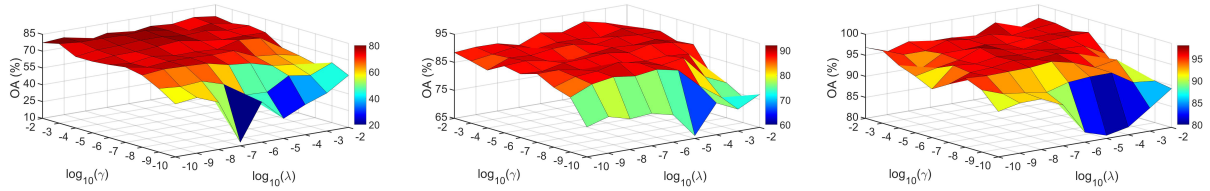
[3] https://engineering.purdue.edu/~biehl/MultiSpec/hyperspectral.html

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TABLE I

NUMBERS OF TRAINING AND TEST SAMPLES OF DIFFERENT CLASSES IN THE THREE TEST DATASETS

Indian Pines				Washington DC				Salinas			
Class	Name	Training	Test	Class	Name	Training	Test	Class	Name	Training	Test
1	Alfafa	5	41	1	Roof	10	3119	1	Weeds_1	10	1999
2	Corn-no till	5	1423	2	Road	10	1780	2	Weed_2	10	3716
3	Corn-min till	5	825	3	Trail	10	1392	3	Fallow	10	1966
4	Corn	5	232	4	Grass	10	1254	4	Fallow plow	10	1384
5	Grass/pasture	5	478	5	Shadow	10	1184	5	Fallow smooth	10	2668
6	Grass/tree	5	725	6	Tree	10	1110	6	Stubble	10	3949
7	Grass/pasture-mowed	5	23					7	Celery	10	3569
8	Hay-windrowed	5	473					8	Grapes	10	11261
9	Oats	5	15					9	Soil	10	6193
10	Soybean-no till	5	967					10	Corn	10	3268
11	Soybean-min till	5	2450					11	Lettuce 4 wk	10	1058
12	Soybean-clean till	5	588					12	Lettuce 5 wk	10	1917
13	Wheat	5	200					13	Lettuce 6 wk	10	906
14	Woods	5	1260					14	Lettuce 7 wk	10	1060
15	Buildings-grass-trees-drives	5	381					15	Vineyard untrained	10	7258
16	Stone-steel towers	5	88					16	Vineyard trellis	10	1797
	Total	80	10169		Total	60	9839		Total	160	53969



(a) (b) (c)

Fig. 6. Effect of parameters λ and γ of the LKSR model in the proposed SBD method. (a) Indian Pines dataset. (b) Washington DC dataset. (c) Salinas dataset.

TABLE II

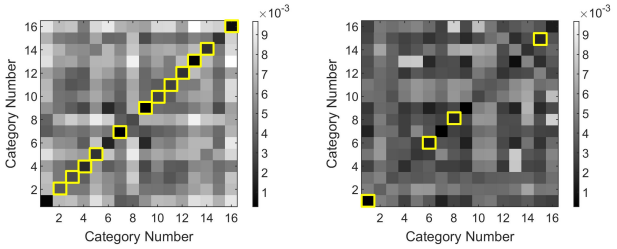
PARAMETERS' SETUP FOR THE PROPOSED SBD METHOD

Parameter	Value	Explanation
$Ratio$	0.65	Ratio of the MNF component number to the original spectral band number
η	0.12	Coefficient in formula (4) to computer the number of superpixels
λ	10^{-6}	Regularization parmeter of the LKSR model
γ	10^{-5}	Kernel parameter of the Log-Euclidean Gaussian kernel

Then, the effect of the coefficient η on the proposed SBD method is analyzed, which is utilized to compute the number of superpixels in (4). In the experiments, η is varied from 0.01 to 0.28 with a step of 0.01, and the other parameters are fixed as values in Table II. As can be seen from Fig. 5, for the three datasets, as η increases, the OA stars go up, then stays a relatively stable trend around a high value, and eventually goes down slightly. The reasons are analyzed as follows. The average size of superpixels in an image is inversely proportional to the coefficient η. The large size of superpixel may consist of pixels from different classes, and the small size of superpixel may lack enough spatial information for HSI classification. Therefore, too small or too large η will reduce the classification accuracy. For the three datasets, the better classification results appear when η is set as values between 0.10 and 0.14. Hence, for the remaining experiments, η is set to 0.12, which is the mid-value of the range. In addition, for the shape-adaptive spatial–spectral block, a simple strategy is adopted to decide the number of the selected similar pixels within the block. In the general case, half the number of most similar pixels in the block are selected to represent the block, where the similarity is measured based on cosine distance C_d. If the number of pixels in this block is less than 50, in order to provide sufficient information for HSI classification, 90% most similar pixels in one block are picked out to represent the block.

Finally, the effect of the regularization parameter λ and the log-Euclidean Gaussian kernel parameter γ is investigated for the LKSR model. In experiments, both parameter λ and parameter γ are varied from 10^{-10} to 10^{-2}. The other parameters are set as in Table II. As shown in Fig. 6, when the values of λ and γ are too small or too large, the image classification accuracies will be reduced to different degrees. The reasons are given as follows. For the regularization parameter λ of the LKSR model, too small a value means that the sparsity of the sparse coefficient is insufficient, while the too large value will result in the inaccuracy of the reconstructed signal. For the log-Euclidean Gaussian kernel parameter γ, too small value will lead to overfitting of the classifier, while too large value cannot well capture the complex structure of nonlinear data to separate it well in mapping space. As can be seen from Fig. 6, for the three datasets, the values of λ and γ for obtaining the better classification results exist the common ranges: λ is in 10^{-7}–10^{-5}, and γ is in 10^{-7}–10^{-3}. Therefore, the mid-values

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(a) (b)

Fig. 7. Normalized distance matrices of (a) SBD and (b) CCM for the Indian Pines dataset.

of the two ranges, i.e., $\lambda = 10^{-6}$ and $\gamma = 10^{-5}$, are used for the remaining experiments.

C. Analysis of the Discriminability Between SBD and CCM

One of the motivations of this article is to fully excavate correlation among different spectral bands by using the SBD for providing more discriminative information. Therefore, in the next experiment, two local correlation features (i.e., SBD and CCM) are compared in the aspect of the discriminability performance for HSI. In the experiment, the Indian Pines dataset is used to construct the normalized distance matrices of SBD and CCM. In the normalized distance matrix, elements on the primary diagonal represent normalized intraclass distances of extracted features, while the other nondiagonal elements represent normalized interclass distances. Ideally, the intraclass should be as small as possible, while the interclass should be as large as possible. The two distance matrices are visualized in Fig. 7. The larger the distance value is, the brighter the corresponding gray block in the figure is.

First, the intraclass distances of SBD and CCM for the same class are compared. For the smaller intraclass distance of the two features, the gray blocks on the diagonal in the corresponding subgraph are marked with yellow boxes. As shown in Fig. 7, the majority of normalized intraclass distances (diagonal elements in the matrix) by the SBD is smaller than that by the CCM, i.e., 12 out of 16 classes. Then, the normalized interclass distances by the SBD and CCM are also compared. Visually, it can be seen that the SBD occupies a higher interclass distance than the CCM, which is reflected by the higher values (brighter image blocks) in the off-diagonal elements. Numerically, we compare all the values in the off-diagonal elements in a statistical manner. For the Indian Pines dataset, the total number of interclass distances for one feature is 120. For the 120 pairs normalized interclass distance of SBD and CCM, 99 interclass distance values of SBD are higher, which accounts for 82.5% of the total. In summary, for pixels within the same class, the average intraclass distance of SBD is less that of CCM; meanwhile, for pixels from different classes, the average interclass distance of SBD is greater than that of CCM. Hence, as a local correlation feature, the SBD can provide more valuable and more discriminative information to improve HSI classification performance.

Note that, by the SBD or the CCM, each feature is represented as an SPD matrix, which lies on a Riemannian manifold space. Thus, it is not appropriate to directly calculate the distance between features of SPD matrices via conventional distance calculation methods on the Euclidean space. Instead, the obtained SBDs and the classical covariance matrices are first mapped into the Euclidean space by the `logm` operator [37]. On the Euclidean space, these mapped matrices are unfolded to vectors, and formula (9) in [29] is then utilized to calculate the normalized distance between any two vectors.

D. Comparison Results of Different Classification Methods

In this section, both classical and state-of-the-art HSI classification methods are used for comparison purposes. Among them, the SVM and MPKCRC methods only exploit the spectral information for HSI classification, while the others take advantage of the spectral–spatial information. For the spectral–spatial information-based classification methods, the EMAP, LBP, and LCMR methods focus on feature extraction for HSI classification. The CCJSR and LCMR methods apply the classical correlation information on HSI classification. The MSSR and SuperPCA methods, respectively, from the multiscale perspective and the dimension reduction perspective, employ the superpixel for HSI classification. The SVM classifier has also been utilized in the EMAP, SuperPCA, and LCMR methods, which is realized by using the LIBSVM software package [41]. For the EMAP method, the range of values for the standard deviation attribute is from 1% to 5% of the mean of the feature, with a step of 1%. The area attribute is extracted according to thresholds of 200, 500, and 1000. The important parameters of the MSSR and LCMR methods are set according to the corresponding recommendations in [19] and [29]. For example, for the MSSR method, the number of multiscales is, respectively, set to 7, 11, and 5 for the Indian Pines, Washington DC, and Salinas datasets. For the LCMR method, the number of MNF components, the size of the fixed window, and the number of selected pixels in the fixed window are, respectively, set to 20, 220, and 25. For the adopting test dataset in [15], [20], and [27], i.e., Indian Pine image, the parameters for the LBP, SuperPCA, and CCJSR methods are set using the default values in the related publications. For the other datasets used in this article (Salinas image and Washington DC image), the parameters for the LBP, SuperPCA, and CCJSR methods are tuned to reach the best results. For the SVM and MPKCRC, the parameters are set to the default values in [5] and [40].

1) Comparison Results of Indian Pines Dataset: The comparison experiment is first performed on the Indian Pines dataset. The final classification maps from the compared methods are illustrated in Fig. 8, and the quantitative results obtained by different methods are reported in Table III. As can be seen, the SVM and MPKCRC methods, which only consider the spectral information, deliver the classification maps with much noise. In contrast, the other methods based on spectral–spatial information significantly improve the classification performance in terms of classification maps and quantitative results. In addition, compared spectral–spatial feature-based classification methods (i.e., EMAP, LBP, LCMR, and SBD), the LCMR and SBD methods that extract

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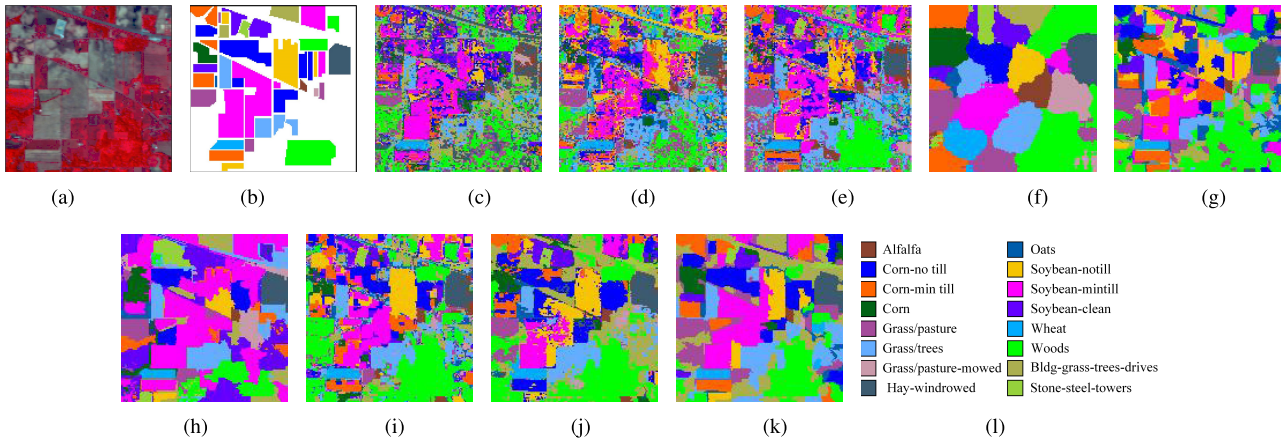


Fig. 8. Indian Pines dataset. (a) False color map. (b) Reference map. Classification maps obtained by different methods: (c) SVM [5], OA = 42.68%; (d) MPKCRC [40], OA = 45.40%; (e) EMAP [17], OA = 61.07%; (f) LBP [15], OA = 68.70%; (g) MSSR [19], OA = 69.67%; (h) SuperPCA [20], OA = 62.60%; (i) CCJSR [27], OA = 67.09%; (j) LCMR [29], OA = 72.73%; and (k) proposed SBD, OA = 76.24%. (l) Class names.

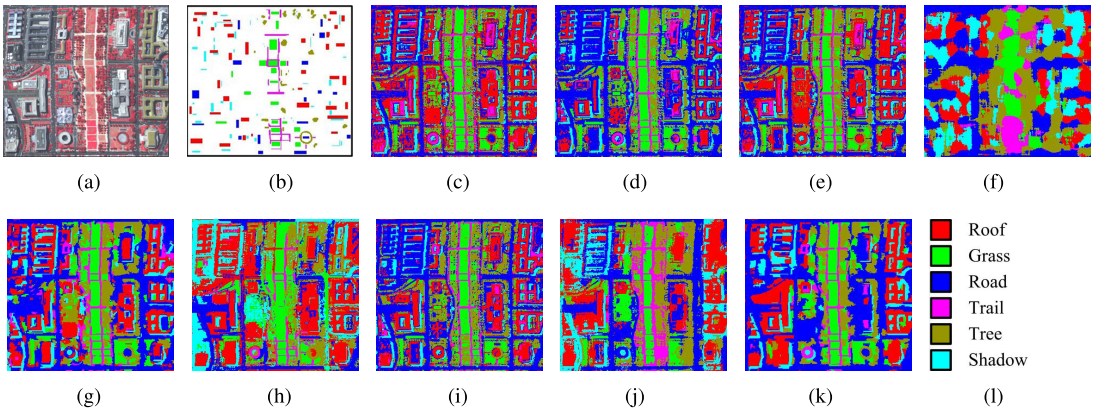


Fig. 9. Washington DC dataset. (a) False color composite map. (b) Reference map. Classification maps obtained by different methods: (c) SVM [5], OA = 84.96%; (d) MPKCRC [40], OA = 83.35%; (e) EMAP [17], OA = 86.78%; (f) LBP [15], OA = 60.81%; (g) MSSR [19], OA = 86.29%; (h) SuperPCA [20], OA = 80.12%; (i) CCJSR [27], OA = 83.38%; (j) LCMR [29], OA = 81.90%; and (k) proposed SBD, OA = 88.63%. (l) Class names.

TABLE III

CLASSIFICATION ACCURACIES ON THE INDIAN PINES DATASET BY THE SVM [5], MPKCRC [40], EMAP [17], LBP [15], MSSR [19], SUPERPCA [20], CCJSR [27], LCMR [29], AND SBD METHODS WITH FIVE TRAINING SAMPLES PER CLASS. NUMBERS IN THE PARENTHESES DENOTE THE STANDARD VARIANCE OF THE ACCURACIES OBTAINED IN REPEATED EXPERIMENTS

Class	SVM	MPKCRC	EMAP	LBP	MSSR	SuperPCA	CCJSR	LCMR	SBD
1	76.83(10.19)	73.65(7.13)	90.48(11.77)	99.75(0.73)	97.80(0.73)	**100(0.00)**	35.81(11.01)	98.78(1.22)	97.56(0.00)
2	29.81(9.67)	27.57(6.35)	41.85(11.16)	45.76(9.86)	49.19(11.04)	40.16(27.17)	56.72(9.17)	63.56(7.26)	**64.88(11.20)**
3	43.09(7.91)	34.17(8.48)	45.86(11.38)	57.15(11.37)	61.53(20.27)	48.32(23.17)	55.38(7.91)	58.59(10.33)	**79.68(9.38)**
4	47.93(14.76)	41.98(11.33)	54.91(12.54)	91.33(9.72)	74.70(12.37)	67.07(27.33)	34.95(9.24)	90.81(10.14)	**96.94(9.18)**
5	60.65(17.75)	67.45(9.67)	72.22(15.85)	75.96(10.45)	72.64(19.86)	79.53(9.18)	**84.87(9.87)**	81.10(18.56)	81.05(13.09)
6	59.10(10.96)	63.87(12.62)	80.07(12.12)	77.60(11.06)	91.56(8.70)	84.14(12.47)	93.74(4.71)	87.12(4.83)	**95.74(5.03)**
7	90.43(5.07)	90.87(6.86)	95.22(3.61)	**100(0.00)**	95.65(0.00)	96.96(1.99)	22.11(6.27)	**100(0.00)**	86.09(28.70)
8	45.56(12.94)	75.43(9.86)	86.34(21.65)	97.56(4.01)	92.07(12.00)	94.37(10.61)	99.76(0.30)	97.67(4.23)	**99.79(0.00)**
9	86.67(15.78)	88.00(9.80)	95.33(14.00)	**100(0.00)**	**100(0.00)**	**100(0.00)**	13.75(3.69)	**100(0.00)**	**100(0.00)**
10	35.84(13.83)	39.96(10.69)	59.04(14.98)	74.23(7.74)	76.36(8.64)	**82.70(3.35)**	59.44(9.54)	70.23(11.34)	80.58(7.01)
11	33.83(13.92)	37.72(16.13)	52.38(9.93)	60.26(4.66)	61.98(11.34)	56.36(30.04)	**74.97(2.95)**	59.15(10.80)	57.30(14.25)
12	29.86(9.71)	33.06(6.82)	53.06(14.36)	55.23(15.56)	55.77(15.53)	48.99(22.75)	50.45(9.02)	68.93(9.69)	**72.52(11.95)**
13	90.55(2.99)	90.55(6.04)	97.95(0.96)	96.95(3.49)	**99.55(0.15)**	99.50(0.00)	84.20(6.45)	99.05(1.03)	99.50(0.00)
14	56.29(13.76)	53.65(8.87)	71.13(9.43)	85.06(8.67)	86.32(11.41)	42.25(17.31)	91.98(5.37)	**93.08(5.03)**	85.71(9.83)
15	29.40(9.38)	28.82(3.90)	69.97(5.24)	**95.69(6.55)**	58.98(13.54)	84.07(15.41)	61.32(17.96)	88.50(14.93)	86.56(9.96)
16	73.64(17.66)	86.25(8.14)	91.25(7.69)	**98.41(1.98)**	97.95(2.02)	90.91(15.64)	93.77(3.48)	96.36(3.79)	96.25(4.25)
OA (%)	42.37(3.76)	44.51(4.41)	60.55(4.35)	69.19(2.48)	69.69(5.34)	61.37(10.01)	65.89(2.86)	72.52(2.18)	**76.18(4.47)**
AA (%)	55.59(4.16)	58.31(1.93)	72.69(4.24)	81.93(1.80)	79.50(3.75)	75.96(4.95)	63.45(1.61)	84.56(1.71)	**86.26(2.71)**
$\kappa \times 100$	36.13(3.91)	38.29(4.22)	55.89(4.75)	65.81(2.69)	65.81(6.08)	56.32(10.43)	63.51(3.09)	70.37(2.36)	**73.24(4.79)**

local correlation features yield better classification results. Furthermore, from the view of using correlation information, the proposed method that can engrave both linear and nonlinear dependence relations is superior to the CCJSR and LCMR methods based on linear correlation information. Moreover, compared the superpixel-based classification methods

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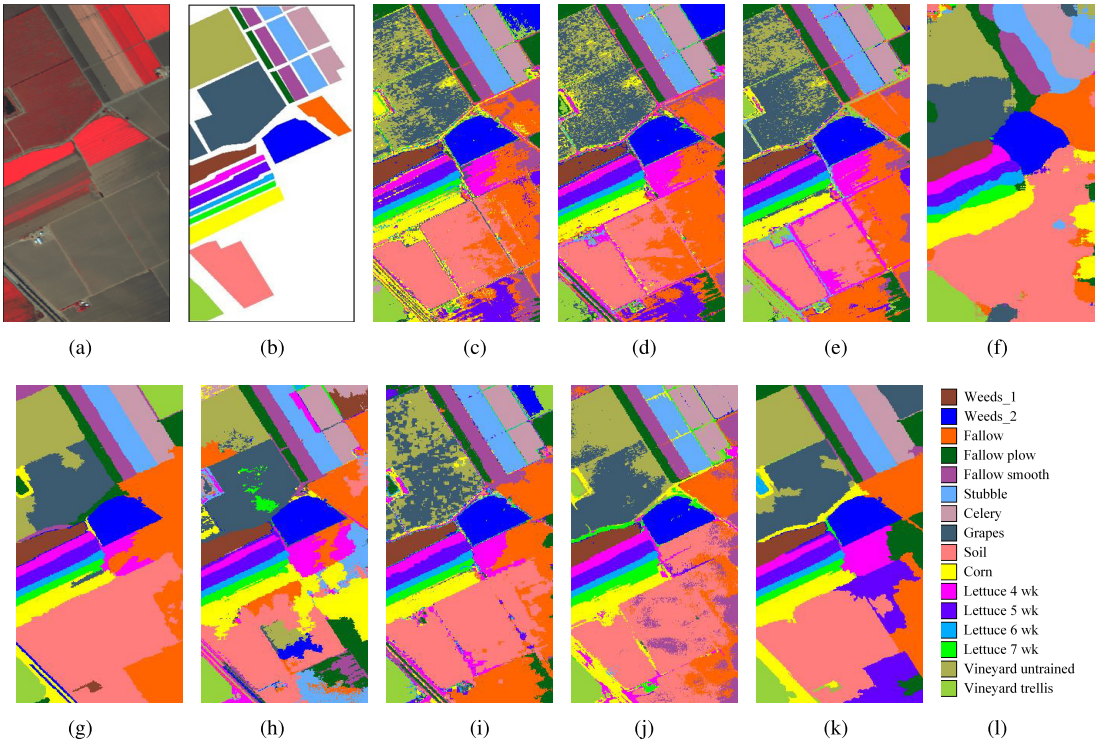


Fig. 10. Salinas dataset. (a) False color composite map. (b) Reference map. Classification maps obtained by different methods: (c) SVM [5], OA = 82.16%; (d) MPKCRC [40], OA = 84.24%; (e) EMAP [17], OA = 87.57%; (f) LBP [15], OA = 90.51%; (g) MSSR [19], OA = 95.68%; (h) SuperPCA [20], OA = 92.05%; (i) CCJSR [27], OA = 86.35%; (j) LCMR [29], OA = 92.74%; and (k) proposed SBD, OA = 96.85%. (l) Class names.

TABLE IV

CLASSIFICATION ACCURACIES ON THE WASHINGTON DC DATASET BY THE SVM [5], MPKCRC [40], EMAP [17], LBP [15], MSSR [19], SUPERPCA [20], CCJSR [27], LCMR [29], AND SBD METHODS WITH TEN TRAINING SAMPLES PER CLASS. NUMBERS IN THE PARENTHESES DENOTE THE STANDARD VARIANCE OF THE ACCURACIES OBTAINED IN REPEATED EXPERIMENTS

Class	SVM	MPKCRC	EMAP	LBP	MSSR	SuperPCA	CCJSR	LCMR	SBD
1	75.72(6.37)	61.02(11.44)	74.39(7.54)	53.53(5.02)	72.21(7.23)	83.45(7.46)	**93.10(2.50)**	78.60(8.33)	79.61(7.89)
2	90.88(8.95)	95.04(4.86)	**97.01(3.94)**	55.29(12.51)	93.47(6.01)	83.03(10.21)	94.09(1.39)	83.90(5.24)	92.47(7.91)
3	85.91(6.71)	86.95(6.02)	88.15(5.68)	75.46(7.65)	85.88(6.98)	66.60(13.34)	64.27(6.36)	78.64(8.28)	**92.41(6.02)**
4	84.96(6.52)	93.95(3.06)	91.84(4.14)	63.27(13.81)	**95.49(1.46)**	88.25(8.28)	74.78(12.35)	89.82(6.36)	92.09(1.04)
5	91.73(1.94)	95.58(1.85)	96.69(1.60)	79.85(5.79)	**96.75(3.03)**	77.95(14.16)	92.93(2.45)	92.12(4.48)	95.46(4.14)
6	97.46(0.81)	97.99(0.88)	92.66(4.58)	49.33(7.22)	95.74(1.98)	82.77(13.11)	**98.51(0.73)**	74.59(12.22)	91.78(1.09)
OA (%)	84.46(1.96)	83.37(4.08)	87.40(2.16)	60.89(1.79)	86.56(2.75)	80.86(3.68)	84.31(2.58)	82.17(3.37)	**88.62(1.72)**
AA (%)	87.77(2.71)	88.42(2.25)	90.12(1.19)	62.79(2.64)	89.92(2.02)	80.34(4.61)	86.28(2.02)	82.95(2.37)	**90.64(1.33)**
$\kappa \times 100$	81.99(2.59)	79.81(4.68)	84.51(2.54)	52.60(2.20)	83.55(3.31)	76.12(4.80)	80.91(2.97)	78.09(3.97)	**85.98(2.07)**

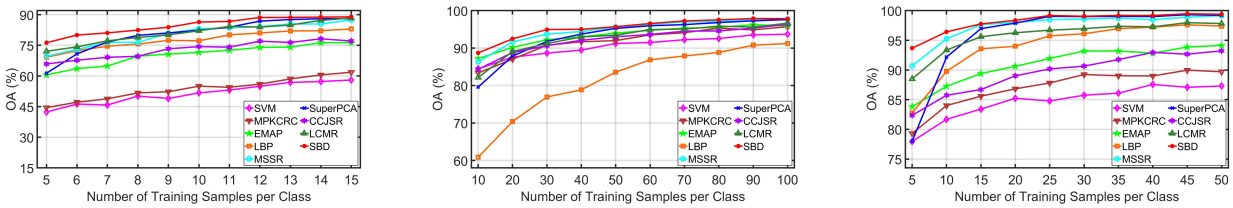


Fig. 11. Effect of the number of training samples on the SVM [5], MPKCRC [40], EMAP [17], LBP [15], MSSR [19], SuperPCA [20], CCJSR [27], LCMR [29], and SBD Methods for (a) Indian Pines dataset, (b) Washington DC dataset, and (c) Salinas dataset.

(i.e., MSSR, SuperPCA, and SBD), the proposed SBD method also achieves more accurate estimation, which means that the SBD as a superpixel-based feature can provide enough spatial–spectral information for the HSI classification. In terms of OA, AA, and kappa, the proposed SBD method obtains the best results. The reason is twofold. First, instead of separately processing the spectral bands or only utilizing linear correlation among spectral bands of the HSI, the SBD can

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TABLE V

CLASSIFICATION ACCURACIES ON THE SALINAS DATASET BY THE SVM [5], MPKCRC [40], EMAP [17], LBP [15], MSSR [19], SUPERPCA [20], CCJSR [27], LCMR [29], AND SBD METHODS WITH TEN TRAINING SAMPLES PER CLASS. NUMBERS IN THE PARENTHESES DENOTE THE STANDARD VARIANCE OF THE ACCURACIES OBTAINED IN REPEATED EXPERIMENTS

Class	SVM	MPKCRC	EMAP	LBP	MSSR	SuperPCA	CCJSR	LCMR	SBD
1	98.64(0.62)	98.10(0.97)	98.08(4.26)	91.71(3.75)	**100(0.00)**	96.29(11.14)	99.61(0.49)	99.97(0.02)	**100(0.00)**
2	97.70(2.68)	97.78(1.91)	98.32(1.50)	98.04(1.97)	99.85(0.10)	99.43(0.57)	99.54(0.91)	91.60(6.30)	**99.96(0.09)**
3	88.48(11.93)	96.10(2.74)	96.44(4.18)	99.69(0.64)	**100(0.00)**	96.84(8.27)	95.92(3.28)	99.56(0.49)	**100(0.00)**
4	96.84(3.88)	99.34(0.32)	99.44(0.47)	90.26(4.12)	99.93(0.00)	94.37(10.34)	96.25(2.38)	**99.99(0.02)**	99.93(0.00)
5	94.28(3.96)	94.00(3.79)	96.50(0.91)	89.50(4.39)	99.31(0.01)	95.89(6.58)	91.76(2.50)	97.04(0.93)	**99.33(0.01)**
6	96.20(3.44)	96.53(1.83)	96.49(3.01)	94.81(2.28)	99.90(0.02)	92.97(6.81)	98.46(3.90)	99.67(0.00)	**99.92(0.00)**
7	99.02(0.62)	98.70(0.95)	98.19(1.03)	91.14(3.81)	**99.86(0.04)**	94.39(6.58)	98.26(0.59)	90.52 (0.36)	99.67(0.36)
8	53.16(11.72)	59.98(7.15)	65.62(17.57)	81.50(10.89)	86.57(11.21)	**93.23(5.46)**	78.29(4.12)	86.91(5.55)	90.52(6.75)
9	97.00(1.73)	97.04(1.61)	95.77(1.20)	98.53(1.57)	**99.88(0.36)**	96.47(3.13)	98.32(0.46)	96.69(1.65)	99.71(0.56)
10	78.56(11.18)	84.54(5.16)	86.38(4.28)	90.86(3.16)	96.10(3.02)	89.82(7.92)	92.15(4.39)	94.30(8.35)	**97.59(0.48)**
11	90.35(5.54)	93.26(3.15)	95.03(1.49)	92.58(7.31)	**100(0.00)**	96.11(3.07)	92.19(1.23)	99.94(0.05)	97.88(1.39)
12	97.61(2.15)	98.97(2.08)	99.97(0.08)	86.19(7.44)	**100(0.00)**	95.28(8.52)	95.17(1.25)	99.41(1.19)	**100(0.00)**
13	98.06(0.55)	98.32(0.69)	98.29(0.76)	83.68(9.59)	97.79(0.00)	97.98(0.34)	77.52(6.98)	**98.41(0.68)**	98.07(0.53)
14	90.06(1.26)	91.17(2.42)	91.92(3.48)	79.76(9.41)	83.56(25.16)	93.89(4.62)	95.21(1.12)	**96.84(1.32)**	94.61(3.65)
15	64.66(6.28)	66.22(7.38)	77.92(8.86)	84.02(6.86)	88.28(7.96)	78.49(12.34)	56.41(6.05)	86.06(7.26)	**91.92(9.26)**
16	90.97(7.32)	86.89(9.07)	96.47(5.66)	**100(0.00)**	97.23(0.02)	92.17(12.05)	99.43(0.29)	93.22(7.03)	97.45(2.98)
OA (%)	81.69(2.70)	84.01(1.94)	87.26(3.13)	89.77(2.36)	95.49(0.93)	92.49(2.08)	85.74(1.63)	93.36(1.81)	**96.43(1.30)**
AA (%)	89.47(1.62)	91.06(1.17)	93.18(1.18)	90.77(1.18)	97.04(1.64)	93.97(1.18)	91.35(0.68)	96.04(1.12)	**97.91(0.58)**
$\kappa \times 100$	79.73(2.93)	82.26(2.14)	85.89(3.61)	88.69(2.57)	94.98(1.04)	91.63(2.31)	84.18(1.77)	92.62(2.01)	**96.03(1.45)**

TABLE VI

AVERAGE RUNNING TIME (SECONDS) OVER TEN REALIZATIONS FOR THE CLASSIFICATION OF THE INDIAN PINES, SALINAS, AND WASHINGTON DC DATASETS BY THE SVM [5], MPKCRC [40], EMAP [17], LBP [15], MSSR [19], SUPERPCA [20], CCJSR [27], LCMR [29], AND SBD METHODS

Class	SVM	MPKCRC	EMAP	LBP	MSSR	SuperPCA	CCJSR	LCMR	SBD
Indian Pines	1.53	1.51	3.32	15.07	3.79	2.67	28.60	7.95	7.81
Washington DC	2.55	2.84	5.67	12.45	25.25	12.38	18.66	32.15	15.32
Salinas	4.94	5.80	9.21	65.63	16.11	25.11	175.83	42.48	11.83

more fully exploit the dependencies among spectral bands to extract more discriminative features. Second, the BD as a local feature is utilized for the superpixel-based block, which can more reasonably use the spatial information for the HSI classification.

2) Comparison Results of the Washington DC Dataset: The second comparative experiments are conducted on the Washington DC dataset. The classification accuracies are listed in Table IV, and the classification maps are shown in Fig. 9. As can be seen from the classification results, different from the Indian Pines dataset, the spectral-based SVM and MPKCRC methods for the Washington DC dataset can produce better classification results. The reason is that the Washington DC image is rich in spatial details, and the spectral-based classification method is suitable for the detail-rich spatial structure. In addition, for the Washington DC dataset, the proposed SBD method still delivers competitive classification results.

3) Comparison Results of the Salinas Dataset: The third comparative experiment is conducted on the Salinas dataset. For the dataset, only about 0.30% of the labeled samples are randomly selected for training and the remaining 99.70% of samples for testing. Therefore, it is very challenging. The classification maps are illustrated in Fig. 10, and the corresponding quantitative values of classification results are tabulated in Table V. As can be seen, although the proportion of the training samples is quite small, the classification accuracy of the proposed method is relatively high. Moreover, compared to the Salinas and Washington DC datasets, the Salinas dataset has many large homogeneous regions, while the Washington DC dataset consists of many heterogeneous regions. Although the completely different spatial structure of datasets, the proposed SBD technique can yield the best classification performance for them. These results indicate that the SBD can overcome the difficulty caused by uneven spatial distribution and provide sufficiently discriminative information for HSI classification.

E. Comparison Analysis of Computing Time

The average computing times over ten realizations on the three datasets are given in Table VI. All the experiments are performed using MATLAB R2018b with a 2.6-GHz CPU and 16-GB memory. As can be seen from Table VI, the SVM and MPKCRC methods run faster than the other compared methods since the original spectral information is directly input the classifiers in the two methods, without any additional spectral–spatial feature extraction process. Compared to the SVM and MPKCRC methods, the EMAP method extracts different types of structure attributes of objects to improve the HSI classification performance. Thus, the running time increased slightly. Besides, the LBP method exploits the local texture feature in the HSI classification, leading to higher computation complexity. For the MSSR, SuperpixelPCA, and SBD methods, they adopt a superpixel segmentation strategy to extract shape-adaptive spatial information, which generally

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achieves better classification accuracy at the cost of higher time consumption. For the spectral–spatial classification methods based on correlation analysis, the SBD method needs much less running time than the CCJSR and LCMR methods. This is because the SBD method calculates local features on superpixel-based blocks, which can deal with pixels in the block simultaneously. Overall, both superpixel segmentation-based and local correlation analysis-based methods always generate better classification performance because of more effective extraction of spectral–spectral features. Among them, the proposed SBD method takes advantage of balancing the classification performance and operation time.

F. Effect of Training Sample Number on Classification Results

Here, the effect of training samples number for the aforementioned methods on three HSI datasets is discussed. Except for the number of training samples, the other parameters for all compared methods are the same as before. For each class in the Indian Pines, Washington DC, and Salinas datasets, labeled samples are randomly selected in varied numbers (from 5 to 15 for Indian Pines, from 10 to 100 for Washington DC, and from 5 to 50 for Salinas) as the training samples and the rest labeled samples for testing. The OA values of different methods, which are averaged over ten runs at different training samples, are shown in Fig. 11. As can be seen, in most of the cases, the classification accuracies for all methods monotonically increase with the growth of training samples number. In contrast, the proposed SBD method consistently outperforms all the compared methods, indicating its superiority in classifying different classes. Moreover, the proposed method has obvious superiority in the case of small training sample sets, especially for Indian Pines and Salinas images. This is reasonable since these two datasets have a lot of homogeneous regions with large sizes, which will provide sufficient spatial information by the superpixel-based BD.

V. Conclusion

In this article, a novel SBD method for HSI classification has been proposed. In this method, the BD that can fully exploit correlation among different spectral bands is employed on HSI classification for the first time. Moreover, the method combines the BD with the superpixel, which can not only more reasonably use the spatial information to construct spatial–spectral feature but also decrease the computation complexity of the method. Compared with the CCM, the SBD can provide much richer and more discriminative information for the classification task. Experiments have been carried out on three widely used hyperspectral datasets. In these experiments, the obtained SBD features are input into an LKSR classifier to yield the final classification maps. Experimental results demonstrate that the proposed SBD method outperforms several well-known classification approaches in terms of qualitative and quantitative performances.

In the future, one focused research direction will be laid on jointly integrating correlation features with other spectral–spatial features to further improve HSI classification accuracy. For example, deep learning, as one of the hottest topics recently, has shown promising performance in extracting deep semantic information. How to effectively excavate complementary characteristics and develop a joint learning mechanism based on correlation analysis and deep learning is still an open question, which is worthy of in-depth exploration.

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Shuzhen Zhang (Member, IEEE) received the M.S. degree from the National University of Defense Technology, Changsha, China, in 2003. She is currently pursuing the Ph.D. degree with the Laboratory of Vision and Image Processing, Hunan University, Changsha, China.

Since December 2013, she has been an Associate Professor with the College of Information Science and Engineering, Jishou University, Jishou, China. Her research interests include sparse representation and remote sensing image processing.



Ting Lu (Member, IEEE) received the B.S. and Ph.D. degrees from Hunan University, Changsha, China, in 2011 and 2017, respectively.

From 2014 to 2015, she was a Visiting Ph.D. Student with the Department of Information Engineering and Computer Science, University of Trento, Trento, Italy, supported by the China Scholarship Council. She is currently an Associate Professor with the College of Electrical and Information Engineering, Hunan University. Her research interests include sparse representation, image fusion, and remote sensing image processing.



Shutao Li (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees from Hunan University, Changsha, China, in 1995, 1997, and 2001, respectively.

In 2001, he joined the College of Electrical and Information Engineering, Hunan University. From May 2001 to October 2001, he was a Research Associate with the Department of Computer Science, The Hong Kong University of Science and Technology, Hong Kong. From November 2002 to November 2003, he was a Post-Doctoral Fellow with the Royal Holloway College, University of London, Egham, U.K. From April 2005 to June 2005, he was a Visiting Professor with the Department of Computer Science, The Hong Kong University of Science and Technology. He is currently a Full Professor with the College of Electrical and Information Engineering, Hunan University. He is also a Chang-Jiang Scholar Professor appointed by the Ministry of Education of China. He has authored or coauthored over 200 refereed papers. His research interests include image processing, pattern recognition, and artificial intelligence.

Dr. Li is also a member of the Editorial Board of the *Information Fusion* and the *Sensing and Imaging*. He won the Second Prize National Natural Science Award of China in 2019 and gained two Second-Grade State Scientific and Technological Progress Awards of China in 2004 and 2006. He is also an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) and the IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT (TIM).



Wei Fu (Member, IEEE) received the B.S. and Ph.D. degrees from Hunan University, Changsha, China, in 2012 and 2019, respectively.

From 2016 to 2017, he was a Visiting Ph.D. Student with the Department of Electrical and Computer Engineering, University of Iceland, Reykjavik, Iceland, supported by the China Scholarship Council. Since 2019, he has been an Assistant Professor with the College of Computer Science and Electronic Engineering, Hunan University. His research interests focus on remote sensing image processing and analysis.

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